QUESTION ONE: ENERGY IN A ZIPLINE SYSTEM

A tourist travels down a zipline from a platform 35 m above the ground.

The combined mass of the tourist and equipment is 85 kg.

The tourist starts from rest.

(a)

Calculate the decrease in gravitational potential energy as the tourist travels from the platform to the ground.

(b)

Just before reaching the ground, the tourist's speed is measured as 22 m s^{-1}.

Calculate the kinetic energy of the tourist.

(c)

Compare your answers to (a) and (b).

Determine the amount of energy that was transferred to other forms of energy.

(d)

Analyse how this "missing" energy was transferred and explain why the law of conservation of energy is still satisfied.

(e) Excellence

A second tourist of mass 105 kg uses the same zipline.

Discuss whether a greater mass means a greater percentage of the original gravitational potential energy will be converted into kinetic energy.

Support your answer using energy concepts.

QUESTION TWO: COOLING HOT CHOCOLATE

A mug contains 0.40 kg of hot chocolate at 82°C.

The specific heat capacity of the hot chocolate is approximately

4200 J kg−1 ◦C−14200~J\,kg^{-1}\,^\circ C^{-1}4200 Jkg−1◦C−1

The drink cools to 52°C.

(a)

Calculate the thermal energy lost by the drink.

(b)

The mug itself warms slightly during the cooling process.

Explain how energy is transferred from the hot chocolate to the mug.

(c)

The room temperature is 18°C.

Analyse why the hot chocolate cools rapidly at first and more slowly later.

(d)

A student says:

> The thermal energy disappears as the drink cools.

Evaluate this statement.

(e) Excellence

Two identical mugs contain identical amounts of hot chocolate.

One mug is covered with a lid immediately.

The other is left uncovered.

Discuss how the total energy transfers differ for the two mugs and explain which mug would cool more slowly.

QUESTION THREE: MELTING ICE

A drinks company adds ice to cans of juice.

A 0.12 kg piece of ice at 0°C melts completely.

The latent heat of fusion of ice is

3.3×105 J kg−13.3\times10^5~J\,kg^{-1}3.3×105 Jkg−1

(a)

Calculate the energy required to completely melt the ice.

(b)

Explain where this energy comes from.

(c)

Analyse why the temperature of the ice remains constant while it melts.

(d)

A student claims:

> Because the temperature does not increase during melting, no energy is being transferred.

Evaluate this statement.

(e) Excellence

Two equal masses of ice are placed in different drinks.

Drink A starts at 25°C.

Drink B starts at 50°C.

Discuss which drink will melt the ice faster and explain your reasoning using thermal energy transfer ideas.

QUESTION FOUR: ELECTRICAL HEATING

An electric heater operates at:

- Voltage = 240 V
- Current = 5.0 A

(a)

Calculate the power output of the heater.

(b)

The heater runs continuously for 3 hours.

Calculate the energy transferred by the heater.

Give your answer in joules.

(c)

Explain how electrical energy is transformed into thermal energy inside the heater.

(d)

A second heater operates at the same voltage but draws 8.0 A.

Compare the power outputs of the two heaters.

(e) Excellence

A student says:

> A heater with greater power is always more efficient.

Discuss whether this statement is correct.

Your answer should clearly distinguish between **power** and **efficiency**.

QUESTION FIVE: HYDROELECTRIC POWER STATION

Water falls through a vertical height of 120 m before reaching a turbine.

Each second, 250 kg of water passes through the turbine.

Assume g=10g=10g=10.

(a)

Calculate the loss of gravitational potential energy of the water each second.

(b)

The station generates electrical energy at a rate of

2.4×105 J/s2.4\times10^5~J/s2.4×105 J/s

Determine the difference between the gravitational energy lost and the electrical energy produced each second.

(c)

Suggest where this energy is transferred.

(d)

Analyse why the hydroelectric station cannot convert all of the gravitational potential energy into electrical energy.

(e) Excellence

An engineer proposes increasing the height of the dam while keeping the water flow rate unchanged.

Discuss how this would affect:

- gravitational potential energy
- electrical energy generation
- energy losses

Use energy concepts to justify your conclusions.